

# Quantum feature selection based on Neutral Atom arrays

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## I. INTRODUCTION

Feature selection (FS) is a crucial step in machine learning pipelines, aiming to identify informative and non-redundant variables that enhance predictive performance while reducing overfitting and computational cost [1, 2]. The task is inherently combinatorial and known to be NP-hard [3, 4], leading classical approaches to rely on heuristics or greedy algorithms.

Quantum computing offers new possibilities for tackling such problems more efficiently, particularly through quantum annealing, variational algorithms, or analog Hamiltonian simulation [5–7]. In this work, we propose a quantum-inspired FS method based on arrays of neutral atoms in Rydberg states, which support tunable long-range interactions and programmable spatial configurations [8–10].

We encode feature relevance via local detunings (based on mutual information with the target), and redundancy through pairwise Rydberg interactions modulated by interatomic distances. These distances are obtained from a multidimensional scaling (MDS) projection of the feature redundancy matrix [11, 12]. The resulting system undergoes adiabatic evolution, ending in a ground state that reflects the optimal feature subset.

Our approach is implemented in simulation using Amazon Braket and evaluated on real-world classification datasets. Compared with classical methods such as Boruta and mutual information ranking [13, 14], our quantum method shows competitive performance, particularly when selecting compact feature subsets—highlighting its potential in high-dimensional settings where classical search becomes intractable.

The rest of the paper is outlined, as follows. Section II reviews the process of feature selection in classical machine learning. Our proposal of a quantum feature selection (QFS) method via neutral atom encoding is presented in section III. Section IV describes an efficient data embedding method based on feature relevance. The used datasets as well as the evaluation procedure are shown in section V, and the achieved results in section VI. The paper ends up with the conclusions of the work and some possibilities for future work in section VIII.

## II. FEATURE SELECTION IN MACHINE LEARNING

Feature selection is a fundamental task in machine learning and data preprocessing, aiming to identify a subset of the most informative variables from a high-dimensional dataset. By retaining only the most relevant features, one can improve model interpretability, reduce overfitting, enhance generalization, and significantly decrease the computational burden during training and inference [2, 15].

*Mutual information* (MI) is one of the most popular metrics to assess feature selection due to its capacity to capture non-linear dependencies between a feature and the target variable. For a feature variable  $X$  and a class label  $Y$ , the mutual information is defined as:

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \left( \frac{p(x, y)}{p(x)p(y)} \right), \quad (1)$$

where  $p(x, y)$  denotes the joint probability distribution of  $X$  and  $Y$ . Features with higher mutual information values are considered more relevant for the classification or regression task at hand [4].

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However, selecting features based solely on individual relevance can lead to redundancy, where multiple selected features carry overlapping information. To address this, modern feature selection techniques consider not only the relevance of features to the target but also the redundancy between pairs of features. This trade-off can be formalized as the following optimization objective:

$$\max_{z \in \{0,1\}^N} \left[ \sum_i I(x_i; y) z_i - \lambda \sum_{i < j} I(x_i; x_j) z_i z_j \right], \quad (2)$$

where  $z_i$  is a binary variable indicating whether feature  $i$  is selected, and  $\lambda$  is a regularization parameter controlling the relevance-redundancy balance. This formulation naturally maps to a *Quadratic Unconstrained Binary Optimization* (QUBO) problem, a class of combinatorial optimization problems that can be tackled using classical heuristics or quantum-enhanced approaches [1, 16].

Recent advances have explored quantum annealing, variational quantum algorithms, and analog quantum simulators to solve QUBO-based feature selection tasks more efficiently, particularly in regimes where exhaustive search is computationally prohibitive [3, 17, 18]. In this work, we exploit the analog simulation capabilities of neutral atom systems to implement a quantum-native feature selection scheme that encodes both mutual information and redundancy into the spatial arrangement and control parameters of a Rydberg atomic array.

### III. NEUTRAL ATOM ENCODING FOR QUANTUM FEATURE SELECTION

Neutral atoms excited to Rydberg states have emerged as a promising hardware platform for quantum computation and simulation, particularly in the context of analog quantum optimization. These systems combine excellent scalability, strong and tunable interactions, and flexible spatial programmability [10, 19–21]. Individual atoms are confined using tightly focused laser beams (optical tweezers) that can be arranged in arbitrary two-dimensional geometries, enabling the construction of programmable interaction graphs. Their internal states are coherently manipulated via laser pulses that drive transitions between the electronic ground state  $|g\rangle$  and a highly excited Rydberg state  $|r\rangle$ , which together form the computational basis of an effective qubit.

Each atom thus behaves as a two-level system, with  $|g\rangle \equiv |0\rangle$  and  $|r\rangle \equiv |1\rangle$ . The time-dependent Hamiltonian for an array of  $N$  such atoms can be written as:

$$H_{\text{Ryd}}(t) = \frac{\Omega(t)}{2} \sum_i \left( e^{i\phi(t)} |g_i\rangle \langle r_i| + \text{h.c.} \right) - \Delta_g(t) \sum_i \hat{n}_i - \Delta_l(t) \sum_i p_i \hat{n}_i + \sum_{i < j} V_{ij} \hat{n}_i \hat{n}_j, \quad (3)$$

where  $\Omega(t)$  and  $\phi(t)$  denote the Rabi frequency and phase of the driving laser field, respectively. The detunings  $\Delta_g(t)$  and  $\Delta_l(t)$  implement global and local energy shifts of the Rydberg level, with  $p_i$  representing site-dependent weights. The last term accounts for van der Waals interactions between atoms simultaneously in the Rydberg state, with strength

$$V_{ij} = \frac{C_6}{|\vec{x}_i - \vec{x}_j|^6} \quad (4)$$

where  $C_6$  is the dispersion coefficient determined by the principal quantum number of the excited state.

A central phenomenon enabling the use of Rydberg atoms for quantum information processing is the *Rydberg blockade*. It arises from the steep distance dependence of the van der Waals interaction. When two atoms are closer than a critical distance  $R_b$ , the interaction energy shift  $V_{ij}$  becomes larger than the energy bandwidth provided by the laser coupling (i.e., the effective Rabi frequency and detuning). As a result, the doubly excited state  $|r_i r_j\rangle$  is shifted out of resonance and cannot be simultaneously populated. This mechanism effectively enforces an exclusion rule: only one excitation can occur within a blockade volume. The blockade radius is given by:

$$R_b = \left( \frac{C_6}{\sqrt{\Omega^2 + \Delta^2}} \right)^{1/6}, \quad (5)$$

which introduces native *hard constraints* based on spatial proximity. This property makes Rydberg systems especially suitable for solving constraint-based problems such as Maximum Independent Set (MIS) [22–24]. In this work, we exploit this physical behavior to encode feature selection as a constrained optimization problem, where each atom represents a feature and the spatial arrangement encodes redundancy constraints.

In comparison to other leading quantum hardware platforms, such as superconducting qubits or trapped ions, neutral atom arrays provide several distinctive advantages. They support system sizes reaching hundreds of individually addressable qubits, offer flexible

and reconfigurable topologies through optical tweezer rearrangement, exhibit strong and tunable interactions, and naturally implement two-dimensional connectivity [25]. State readout is typically performed via fluorescence imaging: atoms remaining in the electronic ground state scatter photons, while those excited to a Rydberg state remain dark. This mechanism enables high-fidelity, single-shot projective measurements with site-level spatial resolution [26–28].

### A. Mapping Feature Selection to the Rydberg Hamiltonian

Feature selection is formulated as the optimization of a cost function combining feature relevance and redundancy. Let  $I_i = I(x_i; y)$  denote the mutual information between feature  $x_i$  and the target variable  $y$ , and  $R_{ij} = I(x_i; x_j)$  the redundancy between feature pairs. The objective is expressed as:

$$Q(x; \alpha) = -\alpha \sum_i I_i x_i + (1 - \alpha) \sum_{i < j} R_{ij} x_i x_j, \quad (6)$$

where  $x_i \in \{0, 1\}$  encodes feature selection, and  $\alpha \in [0, 1]$  balances relevance and redundancy. This function penalizes simultaneous selection of redundant features while favoring those informative with respect to the target.

The mapping is implemented in the final-time Hamiltonian:

$$H_{\text{Ryd}}(T) = -\Delta_l^{\max} \sum_i p_i \hat{n}_i + \sum_{i < j} V_{ij} \hat{n}_i \hat{n}_j, \quad (7)$$

where  $p_i$  encodes normalized feature relevance, and  $V_{ij}$  reflects redundancy via blockade-induced interactions. The resulting energy landscape enforces a natural trade-off between relevance and redundancy.

### B. Adiabatic Evolution Protocol

To prepare the system in the ground state of the target Hamiltonian, we employ an adiabatic evolution strategy. The protocol begins with all atoms in  $|g\rangle$  and interpolates smoothly to the problem Hamiltonian at  $t = T$ . The schedules are designed to maintain adiabaticity while encoding the relevance–redundancy trade-off, as shown in Fig. 1:

- $\Omega(t)$  is ramped up and smoothly decreased to zero, enabling Hilbert space exploration.

- $\Delta_g(t)$  starts from a large negative value to suppress initial excitations and is reduced to zero mid-protocol.
- $\Delta_l(t)$  is activated in the second half, encoding the problem-dependent relevance profile  $\{p_i\}$ .

Rather than explicitly minimizing  $Q(x; \alpha)$ , the system naturally converges to balanced low-energy configurations in which blockade-induced constraints suppress redundant excitations.

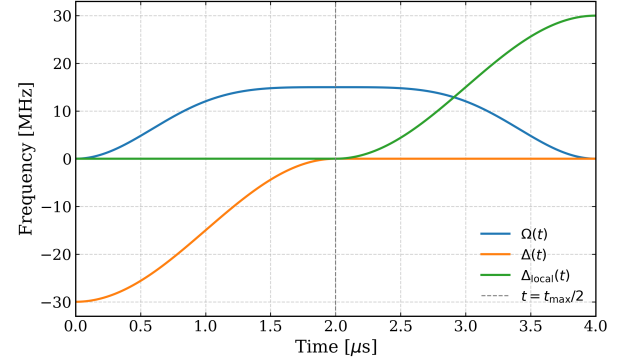


FIG. 1: Driving profiles for selected schedules. The global detuning ensures initialization, the Rabi frequency allows exploration, and the local detunings encode the feature relevance profile.

### C. Simulation Environment

All simulations were carried out using the `braket.ahs` module in the Amazon Braket SDK [29], which provides a high-level interface for defining analog Hamiltonian programs tailored to neutral atom hardware. We employed the Schrödinger solver backend, incorporating programmable local detuning via the `LocalDetuning` class, to encode problem-specific feature relevance. This environment allows precise control of the encoding and efficient benchmarking of our QFS approach.

## IV. EFFICIENT DATA EMBEDDING BASED ON FEATURE IMPORTANCE

To exploit the analog capabilities of neutral atom systems for feature selection, we propose a data embedding strategy that maps the mutual information structure of the dataset into the spatial configuration of an atomic array. This embedding is designed to reflect both feature relevance and pairwise redundancy in a physically meaningful way, consistent with the underlying quantum Hamiltonian.

### A. Mapping Feature Relationships to Physical Distances

The first step is to encode the mutual information between each feature and the target label  $I_i = I(x_i; y)$  into the local detuning amplitudes applied to each atom. These determine the energetic bias for excitation and represent feature importance. Simultaneously, the pairwise mutual information  $R_{ij} = I(x_i; x_j)$  between features is used to define spatial constraints: highly redundant features should be placed closer together to strengthen Rydberg interactions, while weakly correlated features should be kept apart.

This relationship is formalized by constructing a theoretical distance matrix  $D_{ij}$  derived from the redundancy matrix  $R_{ij}$ . For features  $i$  and  $j$ , the raw (unnormalized) distance is computed as:

$$d_{ij} = \left( \frac{1}{R_{ij}} \right)^{1/6} \quad (8)$$

where  $\beta$  is a tunable scaling parameter that regulates how strongly feature importance influences the relative distance. The first term ensures that more redundant features are placed closer, while the second introduces a repulsion effect for highly informative features to avoid overcrowding.

### B. Normalization and Physical Constraints

To embed the redundancy matrix into a neutral-atom layout consistent with hardware capabilities, the computed distances  $d_{ij}$  are normalized to a physically meaningful interval. We define:

$$d_{\min} = \frac{1}{\sqrt{2}} R_b, \quad d_{\max} = 4R_b, \quad (9)$$

where the blockade radius is determined from the final-time parameters as:

$$R_b = \left( \frac{C_6}{\Delta_l^{\max 2}} \right)^{1/6}, \quad (10)$$

being  $C_6$  the van der Waals dispersion coefficient.

At the lower bound  $d_{\min}$ , the interaction term (Eq. 4) is of the same order than the effective driving energy scale ( $0.1 < \Delta_l^{\max}/V_{ij} < 1$ ). This ensures that excitations

of two nearby atoms are energetically off-resonant, effectively implementing a blockade constraint that dominates over local detuning and Rabi coupling. Consequently, features embedded at such separations are strongly penalized when redundantly selected.

In contrast, at the upper bound  $d_{\max}$  the van der Waals interaction decays as  $d^{-6}$ , reaching values several orders of magnitude below ( $\Delta_l^{\max}/V_{ij} \gg 1$ ). In this regime, interatomic interactions are negligible and the excitation probability of each atom is governed almost exclusively by its local detuning amplitude  $p_i$ . Thus, features placed at  $d \sim d_{\max}$  behave effectively independently.

By linearly normalizing the theoretical redundancy-based distance matrix  $d_{ij}$  to  $[d_{\min}, d_{\max}]$ , we obtain the scaled matrix  $D_{ij}^{\text{scaled}}$  that balances physical feasibility and faithful encoding of feature correlations. This procedure ensures that strongly redundant features are placed within the blockade-dominated regime, while weakly correlated features are embedded in the independent-driving regime.

### C. Multidimensional Scaling and Atom Placement

With the normalized distance matrix in hand, we use classical MDS to find a configuration of atomic positions  $\vec{x}_i \in \mathbb{R}^2$  such that the Euclidean distances between atoms approximate  $D_{ij}^{\text{scaled}}$ . Due to the non-Euclidean nature of the input constraints, exact preservation is generally not feasible. Thus, the MDS is run iteratively, and a minimum distance condition is enforced:

$$\min_{i \neq j} \|\vec{x}_i - \vec{x}_j\| \geq R_b \cdot \text{dist\_ratio}, \quad (11)$$

where  $\text{dist\_ratio} = 1/\sqrt{2}$  ensures that all atoms respect the blockade threshold.

If the minimum distance is violated, the interval  $[d_{\min}, d_{\max}]$  is shifted until a configuration satisfying the constraint is obtained. This process guarantees that the resulting arrangement is both physically realizable and optimized according to the data structure (Fig. 2).

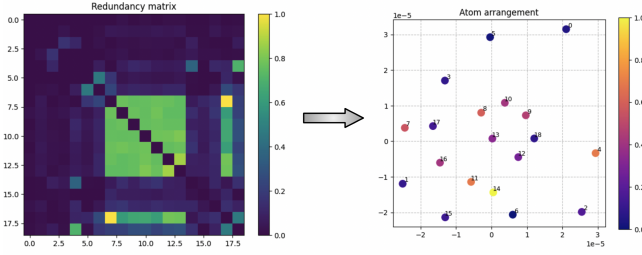


FIG. 2: Mapping of redundancy into actual positions of atoms for one of the datasets. The algorithm is constructed to keep highly-redundant features far away and to maintain closer the non-redundant ones.

#### D. Error Analysis and Robustness

To assess the fidelity of the MDS-based spatial embedding, we compute a relative error matrix:

$$\epsilon_{ij} = \left| \frac{D_{ij}^{\text{scaled}} - \|\vec{x}_i - \vec{x}_j\|}{D_{ij}^{\text{scaled}}} \right|, \quad (12)$$

which quantifies the distortion between the target distance  $D_{ij}^{\text{scaled}}$  (derived from feature redundancy) and the realized spatial distance between atoms  $i$  and  $j$ . This matrix provides a detailed, pairwise assessment of how well the MDS configuration respects the desired geometry.

While the mapping from a non-Euclidean distance matrix to a two-dimensional Euclidean embedding is inherently approximate, we find that the final configurations generally exhibit low relative error and preserve the most critical structural correlations necessary for quantum encoding.

Classical MDS algorithms are known to be sensitive to the initial position of points in the optimization landscape, particularly when the input distance matrix encodes competing or non-metric constraints. To enhance robustness and minimize embedding artifacts due to poor local minima, we perform  $N = 100$  independent MDS runs with different random initializations.

This ensemble-based approach mitigates the effects of initialization variability and ensures that the selected embedding is a robust approximation of the ideal distance structure, consistent with the physical constraints of the Rydberg system.

#### E. Measurement and Solution Extraction

At  $t = T$ , the system is measured in the computational basis. Each atom collapses into either  $|g\rangle$  or  $|r\rangle$ , corresponding to  $x_i = 0$  or  $x_i = 1$ , respectively. Repeated measurements yield a distribution of bitstrings, each representing a candidate feature subset.

To isolate the most informative outcomes, we filter the measurement results by retaining only the top 10% of bitstrings with the lowest energy contributions, as evaluated under the cost function  $Q(x; \alpha)$ . This filtering step discards higher-energy states, focusing the analysis on subsets that best capture the trade-off between feature relevance and redundancy enforced by the dynamics.

#### F. Post-Processing and Alternative Subsets

Directly averaging the Rydberg densities  $\langle \hat{n}_i \rangle$  across the filtered low-energy bitstrings may obscure relevant structures: for instance, two features may individually contribute strongly but belong to mutually exclusive subsets. To address this, we implement a redundancy-aware post-processing algorithm that generates multiple independent alternatives while also identifying the most favorable set for each subset size.

The procedure consists of two main stages:

1. **Generation of alternative subsets.** Starting from the features ranked by their average Rydberg density, we iteratively construct up to a fixed number of candidate subsets (by default, four). At each step, features are added in descending order of density, subject to a compatibility check: if the redundancy between a candidate feature and any already selected feature exceeds a threshold (e.g. 0.7), the feature is excluded. This ensures that each alternative emphasizes complementary, non-redundant information.
2. **Selection of the best set for each cardinality  $k$ .** For every possible subset of size  $k$ , the algorithm evaluates all previously generated alternatives and chooses the  $k$  features with the highest average density among those that are redundancy-compatible. If no alternative can provide  $k$  features due to redundancy constraints, a *fallback strategy* is applied: the top- $k$  features by global density are selected, ignoring redundancy. This guarantees that a valid subset is always produced.

This post-processing stage yields two complementary outputs: a collection of *alternative redundancy-aware*

*subsets*, reflecting diverse balanced trade-offs between relevance and redundancy, and a mapping of *best sets by cardinality*, ensuring that exactly  $k$  features are available for any desired subset size. As a result, instead of a single averaged solution, we obtain a richer landscape of candidate feature sets, increasing both the interpretability and flexibility of the QFS pipeline for downstream machine learning tasks.

## V. PERFORMANCE EVALUATION

### A. Datasets

To evaluate the effectiveness and generality of our QFS framework, we conducted experiments on three publicly available binary classification datasets. These datasets span diverse application domains—telecommunications, socioeconomics, and marketing—and include a mix of numerical and categorical variables.

To enable compatibility with information-theoretic measures such as mutual information and redundancy, all categorical variables were transformed into integer codes via label encoding, and missing or non-numeric entries were removed or imputed when necessary. Numerical features were optionally standardized to zero mean and unit variance to improve comparability. Table I summarizes the dimensionality and variable types of each dataset.

#### a. *Adult Income (UCI)*

[www.archive.ics.uci.edu/dataset/2/adult](http://www.archive.ics.uci.edu/dataset/2/adult)

Also known as the “Census Income” dataset, this benchmark task aims to predict whether an individual earns more than \$50K per year. The dataset originally includes 15 attributes such as age, education, occupation, race, marital status, and hours worked per week, totaling 45,222 cleaned samples after preprocessing.

The data includes 9 categorical features (e.g., `workclass`, `occupation`, `native-country`) and 6 numerical features (e.g., `age`, `capital-gain`, `hours-per-week`). All string-based categorical attributes were label-encoded to obtain discrete values, and missing entries denoted by “?” were removed. This transformation enabled the computation of mutual information and pairwise redundancies on a fully numerical representation.

#### b. *Bank Marketing (UCI)*

[www.archive.ics.uci.edu/dataset/222/bank+marketing](http://www.archive.ics.uci.edu/dataset/222/bank+marketing)

This dataset contains information from 7842 phone-

based marketing campaigns conducted by a Portuguese bank. The target is to predict whether a client subscribed to a term deposit. The original dataset includes 18 features such as job type, education level, month of contact, and previous campaign success.

Of these, 10 are categorical (e.g., `job`, `education`, `poutcome`) and 8 are numerical (e.g., `age`, `balance`, `duration`). The categorical attributes were encoded using label encoding to create integer-valued variables compatible with information-theoretic metrics. The variable ID was removed, and all fields were verified to contain valid and non-missing values.

#### c. *Telco Customer Churn (Kaggle)*

[www.kaggle.com/datasets/blastchar/telco-customer-churn](http://www.kaggle.com/datasets/blastchar/telco-customer-churn)

This dataset contains 7043 records describing customers of a telecommunications company, with the objective of predicting whether they eventually canceled their subscription. The dataset originally contains 21 columns, including customer identifiers, demographic information (e.g., gender, senior status), account tenure, service usage (e.g., internet, streaming, tech support), and billing details.

Out of the 21 features, 17 are categorical—including contract type, payment method, and several binary service flags—and 3 are numerical: `tenure`, `MonthlyCharges`, and `SeniorCitizen`. The variable `TotalCharges`, although numerical, was cast to string in some rows and required conversion and cleaning. The identifier column `customerID` was dropped. After preprocessing, all categorical features were label-encoded for compatibility with mutual information computation.

These datasets were selected to test the performance of QFS across diverse data distributions, class imbalances, and redundancy structures. Their mixed-type nature, common in real-world settings, poses a natural challenge for classical and quantum selection methods alike. Our preprocessing ensures consistency and allows for the direct computation of relevance and redundancy matrices used throughout the pipeline.

### B. Evaluation Metrics and Classification Procedure

To evaluate the predictive performance of the feature subsets selected by our QFS method, we trained a supervised learning model using only the selected features and measured its performance on a hold-out test set. Specifically, we used the XGBoost classifier [30], a widely adopted gradient boosting framework known for its robustness and accuracy in tabular data tasks.

TABLE I: Summary of datasets used in the experiments.

| Dataset              | Samples | Features | Classes |
|----------------------|---------|----------|---------|
| Adult Income         | 48,842  | 14       | 2       |
| Bank Marketing       | 45,211  | 16       | 2       |
| Telco Customer Churn | 7,043   | 19       | 2       |

For each selected feature subset, the following evaluation pipeline was applied:

- The data was split into training and testing subsets.
- The selected features were standardized.
- Hyperparameters of the XGBoost classifier were optimized using a randomized grid search over a predefined hyperparameter space (including `n_estimators`, `learning_rate`, `max_depth`, `subsample`, and others).
- Each hyperparameter configuration was evaluated using 5-fold cross-validation on the training set, using AUC (Area Under the ROC Curve) as the scoring function [31].
- The top five models with the highest cross-validation AUC scores were retrained on the full training data and evaluated on the test set to obtain the following metrics [31]:
  - **AUC (ROC)**: Area under the Receiver Operating Characteristic curve, a robust measure of binary classification performance insensitive to class imbalance.
  - **Precision**: The ratio of true positives to the total number of predicted positives.
  - **Recall**: The ratio of true positives to the total number of actual positives.
- The entire process was repeated across five different random seeds to ensure robustness and account for statistical variance in model training and data splitting.

The final reported results consist of the aggregated distributions of the test AUC, precision, and recall scores across the top-performing hyperparameter configurations and multiple random seeds. This evaluation framework ensures that our method is not only selecting theoretically meaningful features, but also yielding subsets that support strong predictive performance in practical downstream tasks.

### C. Classical Methods: Importance Ranking and Boruta

To benchmark the performance of our QFS approach, we compare it against two widely used classical baselines based on gradient boosting classifiers (XGBoost).

**Importance Ranking:** We train an `XGBClassifier` on the training set, using `logloss` as the evaluation metric. After training, feature importances are extracted directly from the model, where the importance score of a feature corresponds to its average contribution to reducing the loss across all decision splits. The features are then ranked in descending order of importance, yielding a deterministic relevance ordering.

**Boruta:** As a complementary wrapper-based method, we employed the Boruta algorithm [13], implemented with XGBoost as the base estimator. Boruta iteratively compares the importance of each real feature with that of randomly permuted (the so-called shadow) features. A feature is retained only if its importance is consistently higher than the threshold defined by the top 25th percentile of the shadow features. This ensures that only statistically robust and non-random features are selected.

Together, these baselines provide two complementary perspectives: a fast model-driven relevance ranking and a statistically grounded wrapper method, against which we evaluate the redundancy-aware quantum approach.

## VI. RESULTS

In this section, we first illustrate the redundancy-aware atomic arrangements obtained from the multidimensional scaling (MDS) embedding of the redundancy matrices. Subsequently, we compare the predictive performance of QFS against classical baselines using AUC, precision, and recall. To ensure clarity, we unify the performance plots for all three datasets in a single figure.

The MDS algorithm successfully reduces the relative error of the target distance matrices, producing embeddings that remain consistent with the redundancies present in the original datasets. As expected, datasets with lower redundancy exhibit flatter distance matrices, which makes the embedding more challenging: small variations in redundancy must be translated into meaningful spatial separations. Conversely, datasets with stronger redundancy patterns yield more structured embeddings and therefore lower relative distortion.

These results confirm that the MDS-based arrangement reliably encodes redundancy structures into blockade-consistent atomic layouts, minimizing the rel-

TABLE II: Comparison of AUC, precision, and recall for selected feature subsets. Bold AUC values indicate cases where QFS achieves equal or superior performance relative to the classical ranking baseline. Horizontal lines separate the different datasets.

| Dataset | Features | Method  | AUC          | Precision | Recall |
|---------|----------|---------|--------------|-----------|--------|
| Adult   | 2        | QFS     | <b>0.857</b> | 0.737     | 0.396  |
|         |          | Ranking | 0.836        | 0.989     | 0.209  |
| Adult   | 3        | QFS     | 0.868        | 0.691     | 0.456  |
|         |          | Ranking | <b>0.894</b> | 0.781     | 0.523  |
| Adult   | 5        | QFS     | 0.887        | 0.718     | 0.541  |
|         |          | Ranking | <b>0.907</b> | 0.791     | 0.569  |
| Bank    | 2        | QFS     | <b>0.875</b> | 0.772     | 0.481  |
|         |          | Ranking | 0.783        | 0.691     | 0.358  |
| Bank    | 3        | QFS     | <b>0.890</b> | 0.756     | 0.515  |
|         |          | Ranking | 0.888        | 0.756     | 0.511  |
| Bank    | 5        | QFS     | 0.913        | 0.704     | 0.601  |
|         |          | Ranking | <b>0.913</b> | 0.698     | 0.604  |
| Churn   | 2        | QFS     | <b>0.819</b> | 0.665     | 0.382  |
|         |          | Ranking | 0.796        | 0.529     | 0.625  |
| Churn   | 4        | QFS     | <b>0.834</b> | 0.634     | 0.464  |
|         |          | Ranking | 0.815        | 0.612     | 0.525  |
| Churn   | 5        | QFS     | 0.834        | 0.634     | 0.464  |
|         |          | Ranking | <b>0.837</b> | 0.638     | 0.479  |

ative error of the redundancy matrices. While embeddings are more challenging for datasets with limited redundancy, QFS nonetheless produces consistent layouts and competitive predictive performance. In particular, the method seems specially suitable for compact feature subsets, highlighting its potential as a scalable and redundancy-aware alternative to classical feature selection.

## VII. HYPERPARAMETER OPTIMIZATION OF XGBOOST

To ensure the XGBoost classifier achieved optimal performance for our feature selection evaluation, we conducted a hyperparameter optimization using `RandomizedSearchCV` with 5-fold cross-validation. The optimization process sampled 50 configurations from the following parameter distributions:

- **n\_estimators**: {100, 200, 300, 500}
- **learning\_rate**: {0.01, 0.05, 0.1, 0.2}
- **max\_depth**: {3, 5, 7, 10}
- **subsample**: {0.6, 0.8, 1.0}
- **colsample\_bytree**: {0.6, 0.8, 1.0}
- **gamma**: {0, 0.1, 0.5, 1}

- **min\_child\_weight**: {1, 5, 10}

The objective function to be maximized was the mean ROC-AUC across the five validation folds. The best-performing configurations are summarized in Table III.

The results indicate that optimal performance is achieved for moderately deep trees (`max_depth` between 5 and 10), high subsample and column sample ratios, and relatively small learning rates. Interestingly, several of the top configurations use a non-zero `gamma`, suggesting that additional regularization contributes positively to generalization in this context.

To further evaluate the behavior of the quantum approach beyond the smallest subsets, we include results for sets of up to 5 features, as shown in Table II. This allows us to highlight that QFS is particularly competitive in the initial selections, while classical methods may catch up or surpass it as the number of features increases.

The results confirm the suitability of QFS in compact feature subsets, where it often matches or outperforms the classical baseline (e.g., two features in the three datasets, and also four features in Churn). Fig. 3 illustrates that as the subset size increases, classical methods tend to regain a slight advantage, particularly in the Adult dataset. This reinforces the view that QFS is especially effective in the critical early stage of feature selection, providing redundancy-aware choices that remain competitive even as more variables are included.

## VIII. CONCLUSIONS AND FURTHER WORK

This work has proposed a quantum-native framework for feature selection based on analog simulation with neutral atom arrays. By mapping classical information-theoretic quantities into a physically programmable quantum system, we have demonstrated how mutual information and redundancy can be embedded into the spatial layout and driving parameters of a Rydberg atom array, yielding a Hamiltonian whose ground state encodes optimal feature subsets.

We have validated the effectiveness of this approach on three diverse benchmark datasets using rigorous classification procedures. The achieved results have shown that, for medium-scale problems with up to 20 features, the predictive performance of the quantum-selected subsets is competitive with that of classical methods such as Boruta and mutual information ranking. Moreover, QFS has exhibited a remarkable advantage when selecting small subsets of features, achieving similar or better



TABLE III: Top 5 XGBoost configurations based on mean ROC-AUC over 5-fold cross-validation.

| n_estimators | lr   | depth | subs. | cols. | gamma | child_w. | AUC    |
|--------------|------|-------|-------|-------|-------|----------|--------|
| 200          | 0.10 | 5     | 1.0   | 0.8   | 0.5   | 1        | 0.9285 |
| 100          | 0.20 | 7     | 1.0   | 0.6   | 0.1   | 1        | 0.9275 |
| 200          | 0.20 | 3     | 0.8   | 0.6   | 0.5   | 1        | 0.9272 |
| 500          | 0.05 | 5     | 1.0   | 0.8   | 0.5   | 10       | 0.9272 |
| 300          | 0.05 | 10    | 0.8   | 0.6   | 0.5   | 1        | 0.9271 |

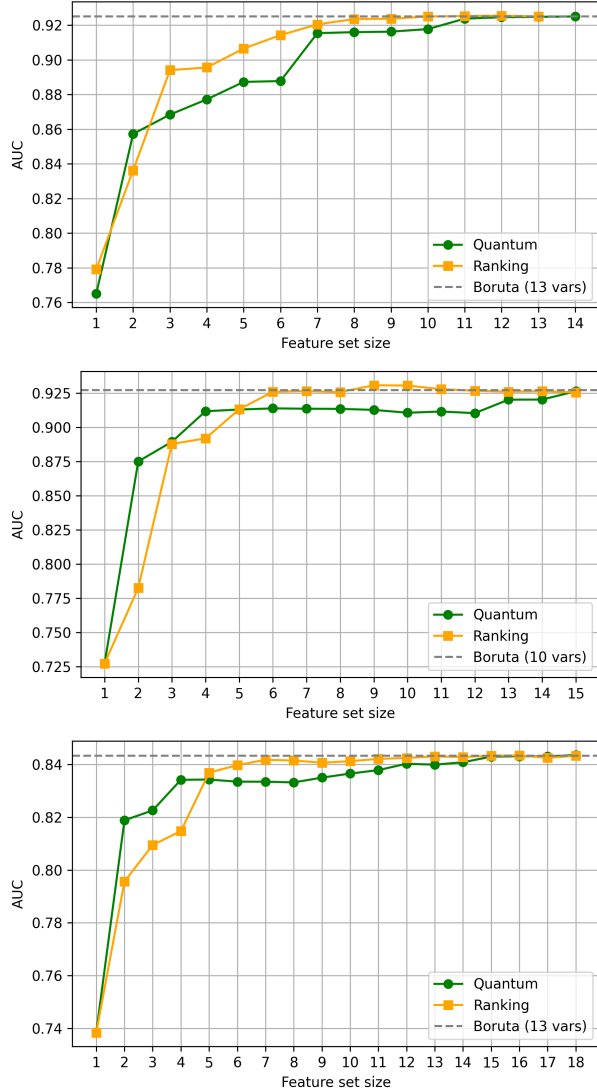


FIG. 3: Comparison of AUC for the three datasets: (top) Adult Income, (middle) Bank Marketing and (bottom) Telco Churn. Results are shown for QFS, Boruta, and mutual information ranking.

classification metrics using fewer variables.

This behavior suggests that QFS may serve as a powerful alternative or complement to classical methods, espe-

cially in high-dimensional settings where exhaustive combinatorial search becomes intractable. The ability of the quantum protocol to encode and optimize over complex interactions via analog constraints offers a new paradigm for intelligent, constraint-aware subset selection.

While the present work has focused on simulations of modest scale, the methodology is designed to be scalable and compatible with near-term hardware implementations on programmable Rydberg platforms. Future directions include applying this framework to larger datasets, integrating local detuning optimization, and exploring unsupervised and semi-supervised approaches.

Overall, our findings highlight the potential of quantum analog computation not only as a tool for solving optimization problems but also as a meaningful contributor to practical machine learning workflows.

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